

NASA Extreme Environment Mission Operations, or NEEMO, is a NASA analog mission that sends groups of astronauts, engineers and scientists to live in the Aquarius underwater laboratory, the world's only undersea research station, for up to three weeks at a time in preparation for future space exploration.

Aquarius is an underwater habitat 3.5 miles (5.6 km) off Key Largo, Florida, in the Florida Keys National Marine Sanctuary. It is deployed on the ocean floor next to deep coral reefs 62 feet (19 m) below the surface.

NASA has used it since 2001 for a series of space exploration simulation missions, usually lasting 7 to 14 days, with space research mainly conducted by international astronauts. The mission had cost about 500 million U.S. dollars. The crew members are called aquanauts (as they live underwater at depth pressure for a period equal to or greater than 24 continuous hours without returning to the surface), and they perform EVAs in the underwater environment. A technique known as saturation diving allows the aquanauts to live and work underwater for days or weeks at a time. After twenty four hours underwater at any depth, the human body becomes saturated with dissolved gas. With saturation diving, divers can accurately predict exactly how much time they need to decompress before returning to the surface. This information limits the risk of decompression sickness. By living in the Aquarius habitat and working at the same depth on the ocean floor, NEEMO crews are able to remain underwater for the duration of their mission.

For NASA, the Aquarius habitat and its surroundings provide a convincing analog for space exploration.

Much like space, the undersea world is a hostile, alien place for humans to live. NEEMO crew members experience some of the same challenges there that they would on a distant asteroid, planet (i.e. Mars) or Moon. During NEEMO missions, the aquanauts are able to simulate living on a spacecraft and test spacewalk techniques for future space missions. Working in space and underwater environments requires extensive planning and

sophisticated equipment. The underwater condition has the additional benefit of allowing NASA to "weight" the aquanauts to simulate different gravity environments.

Until 2012, Aquarius was owned by the National Oceanic and Atmospheric Administration (NOAA) and operated by the National Undersea Research Center (NURC) at the University of North Carolina–Wilmington as a marine biology study base.

Since 2013, Aquarius is owned by Florida International University (FIU). As part of the FIU Marine Education and Research Initiative, the Medina Aquarius Program is dedicated to the study and preservation of marine ecosystems worldwide and is enhancing the scope and impact of FIU on research, educational outreach, technology development, and professional training. At the heart of the program is the Aquarius Reef Base.

Missions

NEEMO 1: October 21–27, 2001

NASA Aquanaut Crew:

Bill Todd , Commander

Michael L. Gernhardt

Michael López-Alegría

Dafydd Williams, CSA

NURC Support Crew:

Mark Hulsbeck

Ryan Snow

NEEMO 2: May 13–20, 2002

NASA Aquanaut Crew:

Michael Fincke , Commander

Daniel M. Tani

Sunita Williams

Marc Reagan

NURC Support Crew:

Thor Dunmire

Ryan Snow

NEEMO 3: July 15–21, 2002

NASA Aquanaut Crew:

Jeffrey Williams , Commander

Gregory Chamitoff

John D. Olivas

Jonathan Dory

NURC Support Crew:

Byron Croker

Michael Smith

NEEMO 4: September 23–27, 2002

NASA Aquanaut Crew:

Scott Kelly , Commander

Paul Hill

Rex Walheim

Jessica Meir

NURC Support Crew:

James Talacek

Ryan Snow

NEEMO 5: June 16–29, 2003

NASA Aquanaut Crew:

Peggy Whitson , Commander

Clayton Anderson

Garrett Reisman

Emma Hwang

NURC Support Crew:

James Talacek

Ryan Snow

NEEMO 6: July 12–21, 2004

NASA Aquanaut Crew:

John Herrington , Commander

Nicholas Patrick

Douglas H. Wheelock

Tara Ruttley

NURC Support Crew:

Craig B. Cooper

Joseph March,

Marc Reagan, Mission Director

NEEMO 7: October 11–21, 2004

NASA Aquanaut Crew:

Robert Thirsk , Commander

Catherine Coleman

Michael R. Barratt

Craig McKinley

NURC Support Crew:

James Talacek

Billy Cooksey

Bill Todd, Mission Director

NEEMO 8: April 20–22, 2005

NASA Aquanaut Crew:

Michael L. Gernhardt , Commander

John D. Olivas

Scott Kelly

Monika Schultz

NURC Support Crew:

Craig B. Cooper

Joseph March

Bill Todd, Mission Director

NEEMO 9: April 3–20, 2006

NASA Aquanaut Crew:

Dafydd Williams , Commander

Nicole P. Stott

Ronald J. Garan Jr.

Timothy J. Broderick , M.D.

NURC Support Crew:

James F. Buckley

Ross Hein

Marc Reagan, Mission Director

NEEMO 10: July 22–28, 2006

NASA Aquanaut Crew:

Koichi Wakata, Commander

Andrew Feustel

Karen L. Nyberg

Karen Kohanowich

NURC Support Crew:

Mark Hulsbeck

Dominic Landucci

Marc Reagan, Mission Director

NEEMO 11: September 16–22, 2006

NASA Aquanaut Crew:

Sandra Magnus, Commander

Timothy Kopra

Robert L. Behnken

Timothy Creamer

NURC Support Crew:

Larry Ward

Roger Garcia

Marc Reagan, Mission Director

NEEMO 12: May 7–18, 2007

NASA Aquanaut Crew:

Heidemarie Stefanyshyn-Piper, Commander

José M. Hernández

Josef Schmid, M.D.

Timothy J. Broderick, M.D.

NURC Support Crew:

Dominic Landucci

James Talacek

Marc Reagan, Mission Director

NEEMO 13: August 6–15, 2007

NASA Aquanaut Crew:

Nicholas Patrick, Commander

Richard R. Arnold

Satoshi Furukawa

Christopher E. Gerty

NURC Support Crew:

James F. Buckley

Dewey Smith

Marc Reagan, Mission Director

NEEMO 14: May 10–23, 2010

NASA Aquanaut Crew:

Chris Hadfield, Commander

Thomas H. Marshburn

Andrew Abercromby

Steve Chappell

Aquarius Reef Base support crew:

James Talacek

Nate Bender

Bill Todd, Mission Director

NEEMO 15: October 20–26, 2011

NASA Aquanaut Crew:

Shannon Walker, Commander

Takuya Onishi

David Saint-Jacques

Steve Squyres

Aquarius Reef Base support crew:

James Talacek

Nate Bender

DeepWorker 2000 submersible crew:

Stanley G. Love

Richard R. Arnold

Michael L. Gernhardt

NEEMO 16: June 11–22, 2012

NASA Aquanaut Crew:

Dorothy Metcalf-Lindenburger, Commander

Kimiya Yui

Timothy Peake

Steve Squyres

Aquarius Reef Base support crew:

James Talacek

Justin Brown

DeepWorker 2000 submersible crew:

Stanley G. Love

Steve Giddings

Serena M. Auñón

Bill Todd

Michael L. Gernhardt

Andrew Abercromby

Steve Chappell

SEATEST II: September 9–13, 2013

Space Environment Analog for Testing EVA Systems and Training

(NEEMO 17) Designation skipped

NASA Aquanaut Crew:

Joseph M. Acaba, Commander

Kate Rubins

Andreas Mogensen

Soichi Noguchi

Aquarius Reef Base support crew:

Mark Hulsbeck

Otto Rutten

NEEMO 18: July 21–29, 2014

NASA Aquanaut Crew:

Akihiko Hoshide, Commander

Jeanette J. Epps

Mark T. Vande Hei

Thomas Pesquet

Professional habitat technicians, Aquarius Reef Base support crew:

James Talacek

Hank Stark (FIU)

NEEMO 19: September 7–13, 2014

NASA Aquanaut Crew:

Randolph Bresnik, commander

Andreas Mogensen, ESA

Jeremy Hansen, CSA

Hervé Stevenin, European Astronaut Centre EVA Training Unit

Aquarius Reef Base support crew:

Mark Hulsbeck

Ryan LaPete

NEEMO 20: July 20 – August 2, 2015

NASA Aquanaut Crew:

Luca Parmitano, ESA, commander

Serena M. Auñón, NASA

David Coan, NASA EVA Management Office engineer

Norishige Kanai, JAXA

Professional habitat technicians, Aquarius Reef Base support crew:

Mark Hulsbeck (FIU)

Sean Moore (FIU)

NEEMO 20 mission objective was to simulate the time delays associated with sending and receiving commands between controllers on Earth and astronauts on Mars. Additional EVAs will simulate working on the surface of an asteroid, and the use of the DeepWorker submersible as an underwater stand-in for the Multi-Mission Space Exploration Vehicle.

NEEMO 21: July 21 – August 5, 2016

The NEEMO 21 mission was scheduled to begin July 18, 2016 and conclude August 3, 2016; however, the mission start was shifted to July 21, 2016 as a result of unfavorable weather conditions.

NASA Aquanaut Crew:

Reid Wiseman, NASA, Commander 1

Megan McArthur, NASA, Commander 2

Marc O'Gríofa

Matthias Maurer, ESA

Noel Du Toit

Dawn Kernagis

Professional habitat technicians, Aquarius Reef Base support crew:

Hank Stark (FIU)

Sean Moore (FIU)

NEEMO 22: June 18–27, 2017

NASA Aquanaut Crew:

Kjell Lindgren, NASA, Commander

Pedro Duque, ESA

Trevor Graff, NASA/Jacobs

Dominic D'Agostino, USF

Professional habitat technicians, Aquarius Reef Base support crew:

Mark Hulsbeck (FIU)

Sean Moore (FIU)

NEEMO 23: June 10–22, 2019

NASA all-female Aquanaut Crew:

Samantha Cristoforetti, ESA, Commander

Jessica Watkins, NASA astronaut candidate

Csilla Ari D'Agostino, a neurobiologist at the University of South Florida

Shirley Pomponi, Marine biologist at Harbor Branch Oceanographic Institute of Florida Atlantic University

Professional habitat technicians, Aquarius Reef Base support crew:

Mark Hulsbeck (FIU)

Tom Horn (FIU)

See also

Continental Shelf Station Two – Undersea research habitat in the Red Sea

SEALAB – Experimental underwater habitats developed by the United States Navy

References

External links

Media related to NEEMO at Wikimedia Commons

Official website

NEEMO missions

Live webcams (subject to mission availability)

Behind the Scenes: NEEMO